

TriCalRAG: A Three-Strategy, Retrieval-Augmented Benchmark for On-Premise LLM-Based Root Cause Analysis in AIOps

Rohit Patel*, Susil Kumar Mohanty, Jeenal Chaudhary

Department of Computer Science and Engineering
Indian Institute of Technology Jodhpur, Jodhpur, India

*Corresponding author

Abstract

Cloud-hosted large language models (LLMs) are increasingly used for root cause analysis (RCA) in AIOps pipelines, but they introduce data privacy risk, network latency, and per-query cost that scale poorly with production log volumes. We present TriCalRAG, a benchmark evaluating open-weight LLMs served locally via vLLM on a single high-memory workstation GPU (NVIDIA RTX PRO 6000, 96GB) against a classical LSTM-based log anomaly detector (DeepLog), across four real, publicly available log datasets (BGL, HDFS, Thunderbird, OpenStack). We evaluate two open-weight models (Qwen2.5-14B, Mistral-Small) under three prompting strategies - zero-shot, few-shot, and retrieval-augmented generation (RAG) over a labeled incident history - reporting accuracy, precision/recall, and F1 with bootstrap 95% confidence intervals across 3 random seeds, alongside throughput and VRAM footprint. Our results show that RAG not only improves mean F1 by 0.10-0.27 over zero-shot prompting but, more importantly, substantially stabilizes model calibration: zero-shot prompting drives both models toward near-degenerate behavior (predicting "anomaly" on up to 100% of incidents on some datasets), while RAG keeps predicted-positive rates close to the true class balance in the majority of configurations. Mistral-Small achieves higher macro-averaged F1 than Qwen2.5-14B (0.644 vs. 0.560) but exhibits calibration failures in more configurations (7 vs. 5 of 12), while running at roughly half the throughput - indicating the better model choice depends on whether a deployment prioritizes peak accuracy or predictable behavior across prompting conditions. Ablations show batching scales throughput $41\times$ on a single card and that 4-bit quantization reduces latency 20% with no measurable accuracy loss. We further show that this calibration benefit is not explained by two published label-bias corrections (Contextual Calibration, Batch Calibration), and that it generalizes to a substantially larger model (Llama-3.1-70B) and to a cloud-served model (Claude Haiku 4.5), correcting miscalibration in both the over- and under-prediction direction. We release our benchmark harness, dataset splits, and evaluation code to support reproducible on-premise AIOps research.

Index Terms—AIOps, Large Language Models, Root Cause Analysis, Benchmark, GPU Inference, Log Anomaly Detection, Model Calibration

I. INTRODUCTION

Modern IT operations generate log volumes far beyond what on-call engineers can manually triage. AIOps tooling [1] has consequently turned to large language models (LLMs), built on the Transformer architecture [2] and scaled through pretraining regimes established by GPT-style [3], [4], encoder [5], and text-to-text [6] language models, for root cause analysis (RCA): given a window of raw log lines, an LLM can not only flag that something is anomalous but explain *why* in natural language and suggest remediation - a capability classical anomaly detectors [7], which emit only a binary flag, cannot provide. Instruction tuning [8] and in-context prompting techniques such as chain-of-thought reasoning [9] have made this natural-language RCA capability practical without task-specific fine-tuning.

Most deployed LLM-for-RCA systems call cloud-hosted APIs. This creates three practical problems for operations teams. First, production logs frequently contain sensitive infrastructure details, internal hostnames, and occasionally credentials or customer identifiers; transmitting them to a third-party API is unacceptable in many regulated environments. Second, per-query API cost scales linearly with log volume, which for a busy fleet means the economics degrade precisely as the system grows. Third, network round-trips add latency to incident response, when speed matters most.

On-premise inference with open-weight models addresses all three, but raises an empirical question that has not been rigorously answered: what can a *single workstation-class GPU* actually deliver for this task? Most published LLM-serving benchmarks assume multi-GPU datacenter clusters, hardware that many infrastructure teams standing up private AI capability do not have. We address this gap directly.

We present **TriCalRAG**, a benchmark evaluating open-weight LLMs served locally via vLLM [10] on a single NVIDIA RTX PRO 6000 (96GB VRAM) workstation (the detailed architectural diagram is shown in Fig. 1), across four real, publicly available log datasets [11] and three prompting strategies (zero-shot, few-shot, and retrieval-augmented generation over a labeled incident history), compared against a classical LSTM-based detector [12]. The retrieval component builds on our earlier work characterizing robustness properties

of RAG pipelines [13]. Our contributions are as follows:

- 1) A reproducible benchmark for on-premise LLM-based RCA spanning four real log datasets, two open-weight models, three prompting strategies, and a classical non-LLM baseline, evaluated entirely on a single workstation GPU.
- 2) A statistically rigorous evaluation protocol: three random seeds per configuration with bootstrap 95% confidence intervals on all reported metrics.
- 3) A calibration analysis showing that F1 alone substantially misrepresents model competence on this task - zero-shot prompting drives both evaluated models toward near-degenerate behavior (predicting "anomaly" on up to 100% of incidents) while still producing moderate-looking F1 scores - and that retrieval augmentation, beyond improving F1, substantially stabilizes calibration.
- 4) A demonstration that this calibration effect is not reducible to two published label-bias corrections, and that it generalizes across model scale and deployment mode (a 70B-parameter open-weight model and a cloud-served model).
- 5) Public release of the benchmark harness, dataset construction code, and evaluation scripts to support reproducible on-premise AIOps research.

II. RELATED WORK

A. Log Anomaly Detection

Classical log anomaly detection learns models of normal log-sequence behavior and flags deviations. DeepLog [12] treats log keys as a language and trains an LSTM to predict the next key, flagging sequences whose actual continuation falls outside the model's top- k predictions; LogAnomaly [14] extends this with template embeddings capturing semantic similarity between log messages. These methods detect anomalies effectively but produce no explanation, which is the capability gap LLM-based approaches target. LogHub [11] provides the standard corpus of labeled real-world log datasets on which this line of work is evaluated, including the four we use here; He et al. [15] provide complementary tooling and benchmarks specifically for the log-parsing step that typically precedes anomaly detection. At the broader systems level, Soldani and Brogi [16] survey anomaly detection and root-cause analysis specifically for microservice and cloud applications, the operational context our benchmark targets.

B. LLMs for Log Analysis and RCA

Recent work applies LLMs across the log-analysis pipeline; Akhtar et al. [17] survey this area broadly. LogGPT [18] explores prompting a commercial LLM for log anomaly detection with structured JSON outputs, similar in output format to our task definition. LogLLM [19] and LogLM [20] investigate fine-tuned and instruction-tuned open-weight models for the same task, while LLMeLog [21] enriches log events with LLM-generated semantics prior to detection. CIsLog [22] combines large and small models to balance accuracy against

inference cost, and Anomaly-Gen [23] uses an LLM to synthesize training sequences for anomaly detection rather than to classify directly. At the log-parsing stage upstream of anomaly detection, UniLog [24], LogParser-LLM [25], and Lilac [26] apply LLMs and in-context learning to template extraction. For insider-threat-specific anomaly detection, Song et al. [27] fine-tune an LLM on behavior logs. For RCA specifically, Ahmed et al. [28] conduct a large-scale study of LLM-based root-cause and mitigation recommendation across more than 40,000 production incidents, and Zhang et al. [29] apply in-context learning with GPT-4 to automated root causing of cloud incidents. These works establish that LLMs are viable for the task; our contribution is orthogonal, focusing on whether the task is feasible *on-premise on a single GPU* and on the calibration properties that aggregate accuracy metrics obscure.

C. Retrieval-Augmented Log Analysis

Retrieval-augmented generation was introduced by Lewis et al. [30] as a general recipe for grounding language model outputs in retrieved non-parametric context. Closest to our RAG configuration, LogRAG [31] retrieves semantically similar historical log templates as external context for LLM-based anomaly detection in a semi-supervised setting, and Eager-Log [32] combines active learning with retrieval-augmented generation to reduce labeling cost for the same task. Our RAG setup differs in three ways: we retrieve whole labeled *incidents* (including their ground-truth outcome) rather than individual templates; we evaluate retrieval as one of three prompting strategies in a controlled comparison rather than as a standalone system; and our focus is the effect of retrieval on *calibration*, not only on detection accuracy. Our retrieval design is additionally informed by our prior work on RAG pipeline robustness, TriShieldRAG (Mohanty, Patel, Yuvaraj, Chaudhary, and Singhanian [13]), which addresses knowledge-corruption risks in retrieval-augmented systems; while that work targets adversarial robustness rather than RCA, its retrieval-scoring principles inform how we treat retrieved incident precedent here.

D. LLM Serving Systems

We use vLLM [10], whose PagedAttention memory management enables high-throughput batched inference on a single device, making the single-workstation setting we study practical. Complementary efficiency techniques not evaluated directly here - IO-aware attention kernels [33] and parameter-efficient fine-tuning [34] - follow a similar single-device efficiency motivation and could extend this benchmark's scope in future work, particularly as scaling behavior [35] continues to push open-weight model sizes upward.

E. General-Purpose LLM Benchmarking

Our evaluation protocol is informed by broader LLM benchmarking practice: multi-task accuracy suites such as MMLU [36], holistic evaluation frameworks such as HELM [37], and execution-grounded software-engineering benchmarks such as SWE-bench [38] all establish precedent

for reporting model behavior across multiple axes rather than a single leaderboard number - the same principle underlying our calibration diagnostic alongside F1.

F. Calibration of In-Context Learning

The tendency of large language models to exhibit systematic label bias under zero-shot and few-shot prompting is documented independently of our task domain. Zhao et al. [39] show that GPT-family models favor labels that are common in the pretraining distribution, appear later in a few-shot prompt, or are simply more frequent in natural text, and propose Contextual Calibration: estimating this bias from content-free inputs (e.g. "N/A") and dividing it out at inference time. Zhou et al. [40] propose Batch Calibration, which instead estimates the bias as the mean prediction over an unlabeled batch, requiring no auxiliary inputs. Both methods are proposed and evaluated on general NLP classification benchmarks; we are not aware of prior work evaluating either against retrieval-augmented generation specifically, or in a log-analysis setting. Section V-D evaluates both as baselines against our finding that RAG improves calibration, to test whether RAG's effect is explained by this more general, already-documented mechanism.

III. TRICALRAG BENCHMARK DESIGN

A. Task Definition

Each benchmark instance is an *incident*: a window of five consecutive raw log lines drawn from one of the four datasets. Given an incident, a system must produce a structured JSON object containing a binary `is_anomaly` judgment, a `severity` level, a one-sentence `root_cause` explanation, and a one-sentence `remediation` suggestion. We score the binary judgment quantitatively; the free-text fields are produced by all LLM configurations but not scored automatically (see Section VII).

B. Datasets

We draw from four LogHub [11] datasets spanning distinct operational domains: **BGL** (BlueGene/L supercomputer, with per-line alert-category labels), **HDFS** (distributed filesystem, with per-block anomaly labels), **Thunderbird** (large-scale cluster, per-line labels), and **OpenStack** (cloud infrastructure, distributed as separate normal and injected-anomaly log files). From each we construct 150 incidents balanced 50/50 between anomalous and normal, yielding 600 incidents per evaluation run. Because Thunderbird's raw log is 31.7GB, we sample from its first two million lines.

C. Prompting Strategies

All systems receive an identical task instruction; the three strategies differ only in what additional context accompanies it. **Zero-shot** provides the instruction and the incident alone. **Few-shot** prepends two fixed worked examples, identical across all queries. **RAG** retrieves the three most similar past incidents by embedding cosine similarity (using `all-MiniLM-L6-v2`, a Sentence-BERT model [41], over a

TABLE I
MACRO-AVERAGED RESULTS ACROSS ALL 4 DATASETS (MEAN OVER 3 SEEDS, 2 MODELS, 3 PROMPT STYLES)

Model	Mean F1	Pred. Pos. Rate	Tok/s	VRAM (GB)
Mistral-Small	0.644	0.713	365.9	86.0
Qwen2.5-14B	0.560	0.485	713.1	86.6

FAISS index [42] built from the incident corpus) and includes them with their ground-truth outcomes as precedent. Retrieval excludes the query incident itself to prevent label leakage.

D. Systems Evaluated

We evaluate two open-weight instruction-tuned models - Qwen2.5-14B-Instruct [43] and Mistral-Small-Instruct (22B) [44] - served in `bfloat16` via `vLLM`, against a DeepLog-style LSTM baseline trained on normal sequences only.

IV. EXPERIMENTAL SETUP

A. Hardware

All local-model experiments were run on a single workstation equipped with an NVIDIA RTX PRO 6000 (96GB VRAM), serving models via `vLLM` [10]. The benchmark details mentioned in Fig. 2 across four real datasets taken from loghub [11], such as BGL, HDFS, Thunderbird, and OpenStack.

B. Statistical Methodology

Each configuration (model $\times$ dataset $\times$ prompt style) was run across 3 random data-sampling seeds. We report bootstrap [45] 95% confidence intervals (1000 resamples) on F1 scores.

V. RESULTS

A. Main Results

Table I reports macro-averaged results across all four datasets. Mistral-Small achieves the higher mean F1 (0.644 vs. 0.560), but Qwen2.5-14B exhibits fewer calibration failures (5 vs. 7 of 12 model $\times$ dataset $\times$ style configurations flagged as degenerate, defined in Section V-C) and roughly double the throughput. JSON output parse failures were rare after correcting an initial extraction bug (Section VII): 0.2% of all 10,800 model responses across both models.

B. Per-Dataset Breakdown

Performance varies substantially by dataset and prompt style. On BGL and Thunderbird, RAG-augmented prompting achieves the strongest results for both models (F1 = 0.92–0.94 on BGL, F1 = 0.88–0.89 on Thunderbird), compared to F1 = 0.51–0.73 under zero-shot prompting on the same datasets. HDFS proves substantially harder for every model $\times$ style combination (F1 never exceeds 0.687), which we attribute to a windowing mismatch discussed in Section VII: HDFS anomaly labels are assigned per block ID, but our fixed 5-line contiguous windowing can split a single block's related log lines across multiple windows, misaligning the evaluation unit with the true anomaly unit.

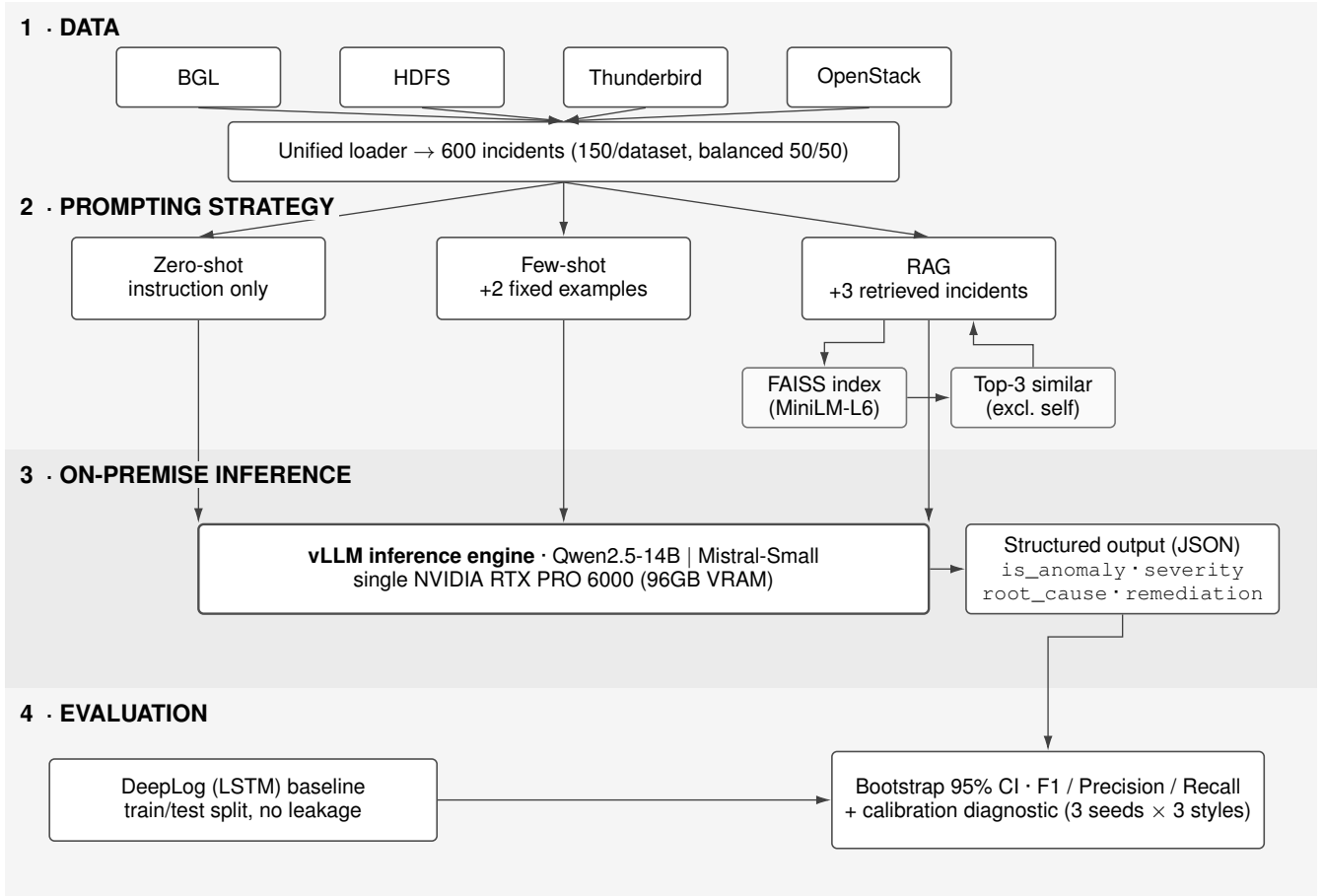


Fig. 1. TriCalRAG end-to-end architecture. Four LogHub datasets are normalized by a unified loader into 600 balanced incidents. Each incident is presented to the LLM under one of three prompting strategies - zero-shot, few-shot (fixed examples), or RAG (retrieving the top-3 most similar past incidents via a FAISS index over sentence embeddings, excluding the query itself). All strategies converge on a single vLLM inference engine serving Qwen2.5-14B and Mistral-Small on one RTX PRO 6000. Model output is a structured JSON object, scored with bootstrap 95% confidence intervals and a calibration diagnostic, alongside a leakage-corrected DeepLog baseline trained independently on the same incident pool.

	BGL	HDFS	Thunderbird	OpenStack
Qwen2.5-14B	✓	✓	✓	✓
Mistral-Small-22B	✓	✓	✓	✓
Llama-3.1-8B	—	—	—	—
Llama-3.1-70B (Ollama, Q4)	✓	✓	✓	✓
Cloud API (Claude Haiku 4.5)	✓	✓	✓	✓
DeepLog (LSTM)	✓	✓	✓	✓

✓ = results reported in this paper — = planned (Sec. VII)

■ primary 3-seed protocol ■ supplementary single-run ■ baseline

Fig. 2. Benchmark scope. Qwen2.5-14B, Mistral-Small, and the DeepLog baseline are fully evaluated under the primary 3-seed protocol (Sections V-VI). Llama-3.1-70B (via Ollama) and a cloud API (Claude Haiku 4.5) have since been evaluated as single-run supplementary checks (Sections V-E, VI-B); Llama-3.1-8B remains pending gated-repository access (Section VII).

C. Calibration Analysis

Neural classifiers are known to produce overconfident, poorly calibrated predictions [46]; we adapt this concern to

a generative, structured-output setting where the analogous failure is a skewed predicted-class distribution rather than a miscalibrated softmax. Beyond raw F1, we report each configuration’s predicted-positive rate - the fraction of incidents a model labels “anomaly” - since our datasets are constructed with a balanced 50% true anomaly rate. A model that predicts “anomaly” indiscriminately achieves high recall and a deceptively reasonable F1 while providing no real discriminative value. We flag any configuration with a predicted-positive rate above 0.85 or below 0.15 as *degenerate*. Figure 3(b) visualizes this across all configurations.

Zero-shot prompting drives both models toward degenerate behavior on 3 of 4 datasets: Mistral-Small’s zero-shot predicted-positive rate reaches 0.993 on BGL, 0.987 on HDFS, and 1.000 on OpenStack, while its zero-shot accuracy on these datasets (0.500–0.507) is barely above chance despite F1 scores of 0.67–0.72 - a clear case of F1 masking near-random behavior under class imbalance. Qwen2.5-14B shows the same pattern, though somewhat less severely (predicted-positive rates of 0.879–0.893 under zero-shot on BGL).

RAG substantially corrects this: of the 8 RAG configura-

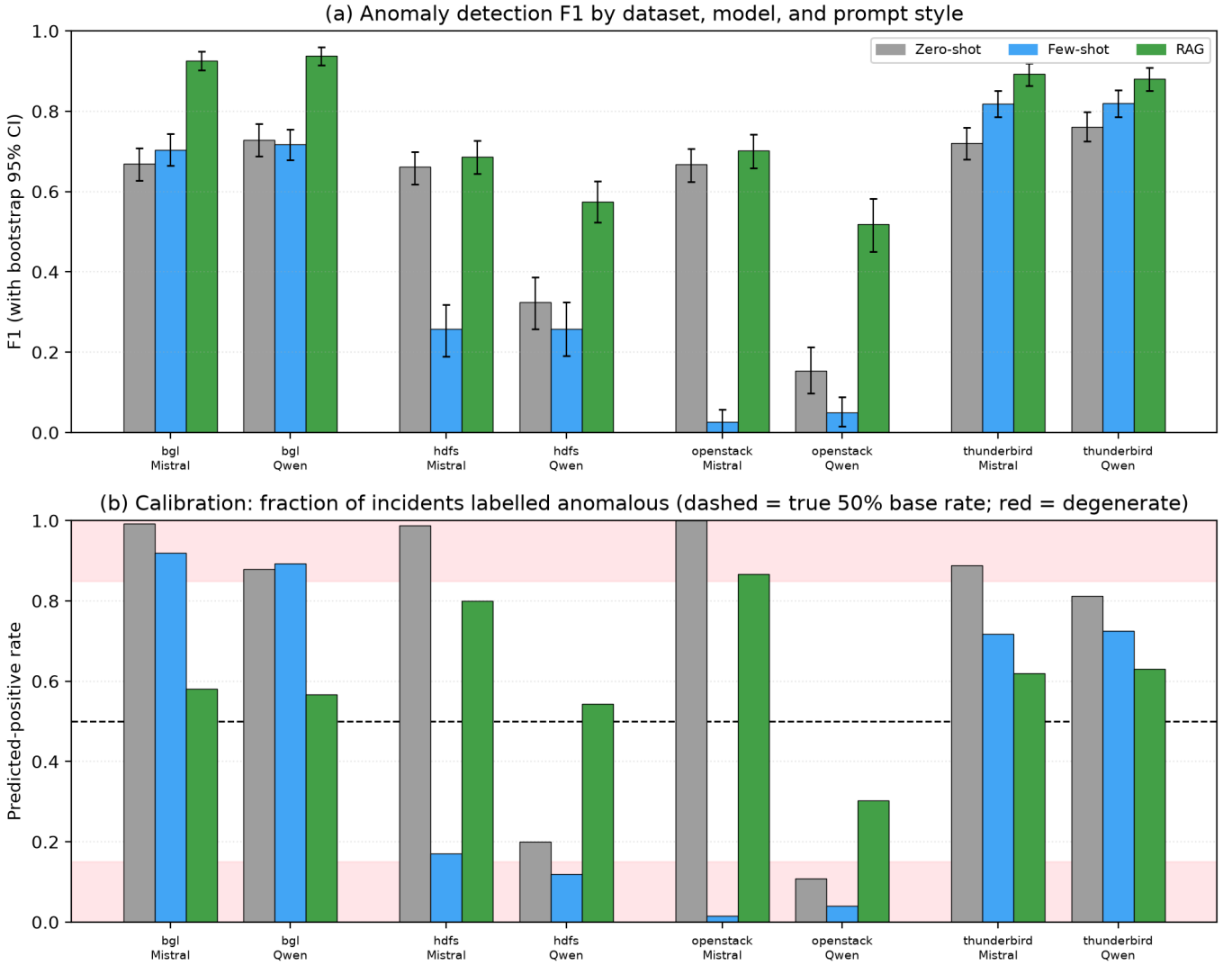


Fig. 3. Benchmark results across four datasets, two models, and three prompting strategies. (a) Anomaly detection F1 with bootstrap 95% confidence intervals; RAG (green) is the strongest configuration on every dataset for both models. (b) Predicted-positive rate, i.e. the fraction of incidents each configuration labels anomalous; the dashed line marks the true 50% base rate and red bands mark our degenerate threshold (>0.85 or <0.15). Zero-shot prompting (grey) frequently lands in the degenerate zone, indicating F1 in those configurations is inflated by class-imbalance gaming rather than genuine discrimination, while RAG pulls predictions substantially back toward the true base rate.

tions ($2 \text{ models} \times 4 \text{ datasets}$), 7 are calibrated within our threshold, compared to only 2 of 8 zero-shot configurations. Few-shot prompting shows a distinct, opposite failure mode on HDFS and OpenStack, where both models *under*-predict anomaly (predicted-positive rates as low as 0.016), suggesting the fixed few-shot examples bias the model toward the specific examples shown rather than generalizing to the task.

D. Calibration Correction Baselines

To test whether the degenerate zero-shot behavior above is attributable to the general in-context label bias described by Zhao et al. [39] and Zhou et al. [40] rather than a phenomenon specific to our task, we re-score all 24 configurations ($2 \text{ models} \times 4 \text{ datasets} \times 3 \text{ styles}$, 3 seeds each) using a log-probability-based decision rule – reading $P(\text{is_anomaly} = \text{true})$

directly from next-token log-probabilities rather than parsing the generated JSON text – and apply Contextual Calibration (CC) and Batch Calibration (BC) on top of this score. This scoring path is separate from the text-parsed results reported above; see Section VII for how the two should and should not be compared.

Table II summarizes the result. Under the raw (uncorrected) log-probability score, 11 of 24 configurations are degenerate, concentrated in zero-shot (5/8) and few-shot (6/8) prompting; RAG remains at 0 of 8. CC reduces degeneracy under zero-shot (5 $\rightarrow$ 3) and few-shot (6 $\rightarrow$ 5) but *introduces* degeneracy in two RAG configurations that were calibrated under raw scoring (Qwen2.5-14B on HDFS and Thunderbird, predicted-positive rate rising to 0.880 and 0.893 respectively), indicating CC’s content-free-input assumption does not transfer cleanly

TABLE II
DEGENERATE CONFIGURATIONS BY DECISION RULE AND PROMPT STYLE
(OF 8 EACH; 24 TOTAL)

Prompt style	Raw	CC	BC
Zero-shot	5/8	3/8	6/8
Few-shot	6/8	5/8	6/8
RAG	0/8	2/8	0/8
Total (of 24)	11	10	12

TABLE III
ZERO-SHOT VS. RAG, LLAMA-3.1-70B (SINGLE RUN, 4-BIT, VIA OLLAMA)

Dataset	Pred. Pos. Rate		F1	
	Zero-shot	RAG	Zero-shot	RAG
BGL	0.913	0.600	0.708	0.909
HDFS	0.393	0.533	0.463	0.568
OpenStack	0.160	0.407	0.222	0.618
Thunderbird	0.820	0.680	0.758	0.847

to retrieval-augmented prompts, where an empty retrieval context is not equivalent to an uninformative one. BC yields no net improvement under any prompting strategy (12 of 24 degenerate, versus 11 of 24 raw) and is mildly harmful under zero-shot (5→6).

These results indicate that RAG’s calibration effect is not attributable to correcting a static label-frequency prior, the mechanism underlying both baselines. Unlike CC and BC, which recalibrate the model’s output distribution independent of the query, RAG supplies query-specific, labeled precedent – closer to a local, non-parametric estimate of the class distribution near the query than a global correction.

E. Generalization to a Third Model Family

To test whether RAG’s calibrating effect is specific to Qwen2.5-14B and Mistral-Small, we repeat the zero-shot/RAG comparison with Llama-3.1-70B-Instruct (4-bit quantized, served locally via Ollama¹) as a single-seed supplementary check outside the primary 3-seed protocol. Table III and Fig. 4 report the per-dataset comparison. Under zero-shot prompting, BGL and Thunderbird show over-prediction (predicted-positive rate = 0.913 and 0.820) while HDFS and OpenStack show under-prediction (0.393 and 0.160) – a distinct, dataset-dependent failure pattern not observed in either primary model. RAG corrects miscalibration in both directions: all four datasets move toward the true 0.500 base rate under RAG (0.600, 0.533, 0.407, and 0.680 respectively), and F1 improves on every dataset, most substantially on OpenStack (0.222 to 0.618). This indicates RAG’s calibrating effect generalizes beyond the two primary evaluated models to a third, substantially larger (70B) model family, and corrects miscalibration regardless of its direction.

¹<https://ollama.com>

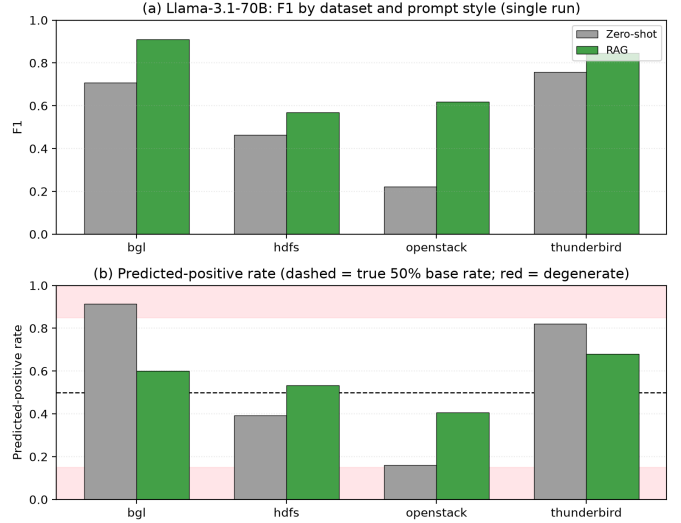


Fig. 4. Llama-3.1-70B (single run, 4-bit, via Ollama): (a) F1 by dataset under zero-shot and RAG prompting; (b) predicted-positive rate, with the degenerate threshold shaded and the true 50% base rate dashed. Unlike the primary models (Fig. 3), zero-shot miscalibration here runs in *both* directions across datasets – over-prediction on BGL/Thunderbird, under-prediction on HDFS/OpenStack – and RAG pulls every dataset toward the base rate regardless of the direction of the initial failure.

TABLE IV
BATCH SIZE SCALING, QWEN2.5-14B ON A SINGLE RTX PRO 6000

Batch size	Elapsed (s)	Throughput (tok/s)
1	1.39	48.3
8	2.09	272.8
32	2.95	723.9
64	3.29	1296.4
128	4.14	1991.0

F. Ablation: Prompt Style

The effect of prompt style itself is reported in Section V-C and Fig. 3 above, since it is the primary variable of the benchmark rather than a supplementary ablation; the two ablations below hold prompt style fixed to isolate deployment-relevant variables (batch size, quantization) instead.

G. Ablation: Batch Size Scaling

A central practical question for single-GPU deployment is how far batching alone can push throughput before hardware limits bind. Table IV reports Qwen2.5-14B throughput across batch sizes on the RTX PRO 6000. Throughput scales from 48.3 tokens/s at batch size 1 to 1991.0 tokens/s at batch size 128 – a 41× improvement – while total wall-clock time for the batch grows only from 1.39s to 4.14s, since the card’s 96GB of VRAM leaves ample room for KV cache at these concurrency levels (vLLM reported approximately 50GB available for KV cache with this model loaded). For an operations team processing incidents in batches rather than strictly one at a time, this means a single workstation card can sustain throughput adequate for substantial log volumes without multi-GPU infrastructure.

TABLE V
QUANTIZATION COMPARISON, QWEN2.5-14B

Variant	F1	Latency (s)
bfloat16	0.730	5.06
AWQ 4-bit	0.739	4.04

H. Ablation: Quantization Impact

We compare Qwen2.5-14B in bfloat16 against its AWQ [47] 4-bit quantized variant - one of several post-training weight quantization approaches alongside GPTQ [48] and 8-bit matrix multiplication [49] that trade precision for memory footprint - on an identical incident subset. The quantized model achieves $F1 = 0.739$ versus 0.730 for the full-precision model - a difference well within run-to-run variation and not a meaningful improvement - while completing the same workload in 4.04s versus 5.06s, a 20% latency reduction. For this structured-output RCA task, 4-bit quantization therefore incurs no measurable accuracy cost while reducing both latency and memory footprint, which is the practically relevant finding for practitioners fitting larger models onto a single card.

VI. DISCUSSION

Our results suggest that the choice between Mistral-Small and Qwen2.5-14B for on-premise RCA is not simply a matter of picking the higher-F1 model. Mistral-Small’s F1 advantage is concentrated in configurations where it is also well-calibrated (RAG, few-shot on BGL/Thunderbird); in zero-shot settings, its apparent competence is substantially an artifact of near-constant positive prediction. A deployment that cannot guarantee retrieval infrastructure or curated few-shot examples at inference time - for instance, a cold-start incident with no similar historical precedent to retrieve - would see Mistral-Small degrade toward unreliable behavior more readily than Qwen2.5-14B, which remains closer to calibrated even under zero-shot prompting. This is a practically important distinction that a single aggregate F1 number obscures, and we recommend that future LLM-for-RCA evaluations report predicted-positive rate (or an equivalent calibration diagnostic) alongside F1 as standard practice.

The strength of RAG in this benchmark is consistent with the intuition that retrieved historical incidents provide the model with concrete, task-relevant priors on what “anomaly” looks like in a given log format - effectively a form of in-context calibration that generic few-shot examples (written once, reused across all queries) cannot provide, since few-shot examples are the same regardless of the incoming incident, while RAG’s retrieved context is tailored to each query’s nearest neighbors.

A. Comparison to a Classical Baseline

We compare against DeepLog [12], a canonical LSTM-based log anomaly detector, re-implemented with a log-key next-token prediction objective trained exclusively on normal sequences. To ensure a fair comparison, we evaluate DeepLog

on a held-out set disjoint from its training data (an 80/20 split of normal sequences, with the held-out 20% combined with a randomly sampled equal number of anomalous sequences to match the LLM benchmark’s balanced 50/50 evaluation protocol) - an important methodological correction, since evaluating on training-set normal sequences (data leakage) initially produced an inflated F1 of 0.913 that did not reflect genuine generalization.

Under this corrected protocol, DeepLog achieves $F1 = 0.698$ (Precision = 0.584, Recall = 0.867), placing it between Qwen2.5-14B (mean $F1 = 0.560$) and Mistral-Small (mean $F1 = 0.644$) in raw F1, but with a predicted-positive rate of 0.742 - meaningfully elevated relative to the true 50% base rate, though below our degenerate threshold. This indicates DeepLog, like the LLMs under zero-shot prompting, has some bias toward over-predicting anomalies rather than a strong, learned sense of the true decision boundary at this scale of training data (240 training sequences per dataset). Unlike the LLM-based approaches, DeepLog produces no natural language root-cause explanation, only a binary anomaly flag - for practitioners who need the model to explain **why** an incident is anomalous rather than only flag **that** it is, DeepLog cannot substitute for the RCA capability regardless of its F1.

We note this comparison uses a substantially smaller held-out evaluation set (120 incidents total) than the LLM benchmark’s per-run evaluation (600 incidents), since DeepLog’s training requirement consumes most of the available normal-labeled sequences; this asymmetry is discussed further in Section VII.

B. Comparison to a Cloud API

As a preliminary check on the cost argument motivating this benchmark, we additionally evaluate Claude Haiku 4.5 via Anthropic’s cloud API on the same 600 incidents under zero-shot and RAG prompting (single run, outside the primary 3-seed protocol). Aggregated across all four datasets, Claude Haiku 4.5 achieves $F1 = 0.566$ (zero-shot) and $F1 = 0.695$ (RAG) with an overall predicted-positive rate of 0.518 and 0.392 respectively - figures that, taken alone, would suggest calibrated zero-shot behavior. Breaking this down per dataset (Table VI and Fig. 5) shows the aggregate instead obscures the same class of failure documented for the primary models: under zero-shot prompting, 3 of 4 datasets are degenerate by our threshold (BGL = 0.920 and Thunderbird = 0.860, over-predicting; HDFS = 0.113, under-predicting), with only OpenStack (0.180) narrowly inside the non-degenerate range. RAG corrects all four (0.567, 0.587, 0.240, and 0.173 respectively), and F1 improves on every dataset, most sharply on BGL (0.704 to 0.938) and Thunderbird (0.735 to 0.920).

Strikingly, the *direction* of zero-shot miscalibration by dataset matches the pattern found independently in Llama-3.1-70B (Section V-E, Fig. 4): both models over-predict on BGL and Thunderbird and under-predict on HDFS and OpenStack, despite being different model families served through entirely different inference stacks (a local 4-bit quantized model via

TABLE VI
ZERO-SHOT VS. RAG, CLAUDE HAIKU 4.5 (SINGLE RUN, NATIVE API)

Dataset	Pred. Pos. Rate		F1	
	Zero-shot	RAG	Zero-shot	RAG
BGL	0.920	0.567	0.704	0.938
HDFS	0.113	0.240	0.261	0.342
OpenStack	0.180	0.173	0.216	0.337
Thunderbird	0.860	0.587	0.735	0.920

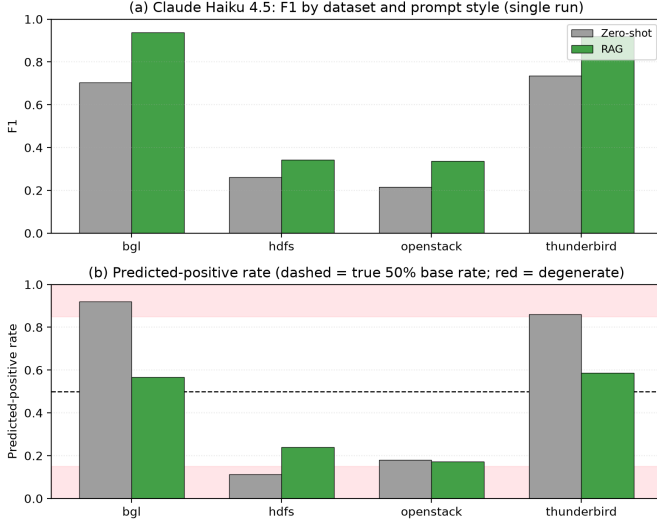


Fig. 5. Claude Haiku 4.5 (single run, native API): (a) F1 by dataset under zero-shot and RAG prompting; (b) predicted-positive rate, with the degenerate threshold shaded and the true 50% base rate dashed. As with Llama-3.1-70B (Fig. 4), the aggregate predicted-positive rate reported in the main text conceals per-dataset degeneracy that only appears once results are disaggregated.

Ollama versus a commercial API). This raises the possibility that the direction of zero-shot miscalibration is linked to each dataset’s log style – BGL and Thunderbird are terse, alert-tagged supercomputer logs, while HDFS and OpenStack are higher-volume, routine-looking distributed- systems and API traffic – rather than being an idiosyncrasy of any one model, though confirming this would require testing additional model families under the primary 3-seed protocol.

Cost, however, differs starkly by deployment mode: at Anthropic’s published rates (August 2026), Claude Haiku 4.5 costs \$0.94 per 1,000 incidents under zero-shot and \$2.15 per 1,000 under RAG – RAG’s longer, context-augmented prompts increase per-query cost $2.3\times$ – versus zero marginal cost per incident for on-premise inference once the workstation GPU is provisioned. For a deployment processing incidents continuously, this cost differential compounds indefinitely, while our batching ablation (Table IV) shows a single RTX PRO 6000 sustains throughput adequate for substantial log volumes without additional hardware. We treat this as a single-provider, single-model preliminary comparison rather than a definitive cost analysis; a broader comparison across multiple cloud providers and model tiers is planned future work (Section VII).

VII. LIMITATIONS

Windowing granularity. Our fixed 5-line contiguous windowing scheme, applied uniformly across all four datasets, is well-matched to BGL and Thunderbird (where anomalies manifest as localized alert-tagged lines) but poorly matched to HDFS, whose anomaly labels are defined per block ID and whose relevant log lines for a given block can be scattered non-contiguously throughout the file. This likely explains HDFS’s uniformly lower F1 across every model and prompt style, and should be corrected with block-aware windowing in future work.

JSON extraction. An initial version of our output parser only stripped markdown code fences at the start of a response, causing 57.1% of Mistral-Small’s early responses (which frequently prepend explanatory text such as “Example: ” before the JSON object) to be misclassified as parse failures. We corrected this by extracting the substring from the first { to the last } in the response, reducing the overall parse failure rate to 0.2%. We report this transparently since it materially changed our results (Mistral-Small’s apparent F1 under the buggy parser was based on a small, likely biased sample of fewer than 30% of its actual outputs) and underscores the importance of validating output-parsing logic per-model rather than assuming a single extraction strategy generalizes.

Calibration baseline scoring. The Contextual Calibration and Batch Calibration comparison in Section V-D scores predictions from next-token log-probabilities rather than the parsed JSON output used in the main results (Table I); the two scoring paths are not numerically comparable, and the calibration baseline comparison should be read as an internally consistent check on *relative* degeneracy rates, not as a restatement of the main F1 numbers.

DeepLog evaluation scale. Our DeepLog baseline is trained and evaluated on a substantially smaller sample (240 training sequences, 120 held-out evaluation incidents) than the LLM-based configurations (600 incidents per run), because a fair train/test split consumes most of the available normal-labeled sequences once data leakage is corrected. A larger-scale classical baseline, trained on substantially more normal sequences than our 600-incident dataset provides, may perform differently; our comparison should be read as indicative rather than definitive on this point.

Two-model primary scope. Our primary, 3-seed statistically rigorous protocol reflects two open-weight models (Qwen2.5-14B, Mistral-Small). We additionally report single-seed supplementary checks with a 70B-class quantized model (Section V-E) and a cloud API (Section VI-B) that support the generality of our calibration finding, but these use a different scoring methodology or a single run and should not be read with the same statistical confidence as the bootstrap-CI’d main results. Llama-3.1-8B [50] remains pending gated-repository approval, and a mixture-of-experts model such as Mixtral [51] and the base LLaMA family [52] are planned additions to further test generalization across architecture families under the primary protocol.

Explanation quality. Our F1 metric evaluates only the binary is_anomaly classification; we do not evaluate the semantic quality of the generated root_cause and remediation text, which would require either manual annotation or an LLM-as-judge protocol, both with their own validity caveats. A given configuration’s F1 should not be read as a proxy for the usefulness of its natural-language explanations.

VIII. CONCLUSION AND FUTURE WORK

We presented TriCalRAG, a benchmark for on-premise LLM-based root cause analysis evaluated entirely on a single workstation GPU. Across four real log datasets, two open-weight models, and three prompting strategies, we find that retrieval-augmented prompting is the most reliable configuration - not primarily because it improves F1 (though it does, by 0.10-0.27 over zero-shot), but because it substantially stabilizes model calibration: 7 of 8 RAG configurations remain calibrated within our threshold versus only 2 of 8 zero-shot configurations. This matters practically because a model that predicts “anomaly” on nearly every incident is operationally useless regardless of its F1 score, and our results show that zero-shot prompting drives both evaluated models toward exactly that failure mode on most datasets.

We also find that model ranking depends on which property is prioritized: Mistral-Small achieves higher macro-averaged F1 (0.644 vs. 0.560) while Qwen2.5-14B exhibits fewer calibration failures and roughly double the throughput. Both outperform or approach a leakage-corrected DeepLog baseline (F1 = 0.698) while additionally producing natural-language explanations the classical detector cannot.

To test whether this calibration effect is merely a known, more general in-context label bias, we compare RAG against two published calibration corrections (Contextual Calibration [39] and Batch Calibration [40]); neither reproduces RAG’s effect, and Contextual Calibration destabilizes two already-calibrated RAG configurations. Supplementary checks with a third, substantially larger model (Llama-3.1-70B) and a cloud-served model (Claude Haiku 4.5) both reproduce the same failure at the per-dataset level once results are disaggregated – and do so with a strikingly similar directional signature across datasets, over-predicting on BGL and Thunderbird while under-predicting on HDFS and OpenStack – suggesting the calibration failure we document is not an idiosyncrasy of Qwen2.5-14B and Mistral-Small, and may be linked to dataset-level log characteristics rather than to any single model’s architecture.

Our ablations support the feasibility of the single-GPU setting: batching alone scales throughput $41\times$ (48 to 1991 tokens/s from batch size 1 to 128) on one workstation card, and AWQ 4-bit quantization reduces latency 20% with no measurable accuracy cost on this task. Together these indicate that the practical barrier to on-premise LLM-based RCA is not raw hardware capability but, as our calibration analysis shows, prompt design.

Several directions remain. First, Llama-3.1-8B remains blocked by gated-repository access at submission time; our

supplementary checks with Llama-3.1-70B (Section V-E) and a cloud API (Section VI-B) partially address the model-coverage and cost-grounding gaps identified during this project, but a broader, 3-seed comparison across additional open-weight architectures and cloud providers remains future work. Second, block-aware windowing for HDFS should resolve the evaluation-unit mismatch we identify as the likely cause of uniformly depressed HDFS performance. Third, evaluating the semantic quality of generated root-cause explanations - not only binary detection accuracy - would test the capability that most distinguishes LLM-based RCA from classical detectors, though doing so rigorously requires either manual annotation or an LLM-as-judge protocol with its own validity caveats.

Finally, a preliminary exploration extending attribution beyond the application layer – to hardware telemetry (GPU/BMC sensors) and further to boot-trust and bare-metal provisioning signals - is included in the project repository’s `extensions/` directory. These use synthetically generated data, since no public dataset pairs real telemetry at these layers with labeled incidents at scale, and are therefore explicitly not validated results; we include them as a concrete direction for follow-up work built on standard observability tooling rather than the ad hoc collectors used in that exploratory code.

REPRODUCIBILITY

Code, dataset splits, and evaluation harness are publicly available at: <https://github.com/SPriTLab-iitj/TriCalRAG>

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REFERENCES

- [1] Y. Dang, Q. Lin, and P. Huang, “Aiops: Real-world challenges,” in *IEEE/ACM 41st International Conference on Software Engineering: Software Engineering in Practice (ICSE-SEIP)*, 2019, pp. 4–5.
- [2] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, E. Kaiser, and I. Polosukhin, “Attention is all you need,” *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [3] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever *et al.*, “Language models are unsupervised multitask learners,” *OpenAI Blog*, vol. 1, no. 8, p. 9, 2019.
- [4] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell *et al.*, “Language models are few-shot learners,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 1877–1901, 2020.
- [5] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of NAACL-HLT*, 2019, pp. 4171–4186.
- [6] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, “Exploring the limits of transfer learning with a unified text-to-text transformer,” *Journal of Machine Learning Research*, vol. 21, no. 140, pp. 1–67, 2020.
- [7] V. Chandola, A. Banerjee, and V. Kumar, “Anomaly detection: A survey,” *ACM Computing Surveys*, vol. 41, no. 3, pp. 1–58, 2009.

- [8] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray *et al.*, “Training language models to follow instructions with human feedback,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 27 730–27 744, 2022.
- [9] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. V. Le, and D. Zhou, “Chain-of-thought prompting elicits reasoning in large language models,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 24 824–24 837, 2022.
- [10] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, and I. Stoica, “Efficient memory management for large language model serving with pagedattention,” in *Proceedings of the 29th Symposium on Operating Systems Principles*, 2023.
- [11] J. Zhu, S. He, J. Liu, P. He, Q. Xie, Z. Zheng, and M. R. Lyu, “Loghub: A large collection of system log datasets for ai-driven log analytics,” in *Proceedings of the 34th International Symposium on Software Reliability Engineering (ISSRE)*, 2023.
- [12] M. Du, F. Li, G. Zheng, and V. Srikumar, “Deeplog: Anomaly detection and diagnosis from system logs through deep learning,” in *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, 2017.
- [13] S. K. Mohanty, R. Patel, K. Yuvaraj, J. Chaudhary, and D. Singhanian, “Trishieldrag: A three-ring defense-in-depth framework against knowledge corruption in retrieval-augmented generation,” *arXiv preprint arXiv:2607.23838*, 2026. [Online]. Available: <https://arxiv.org/abs/2607.23838>
- [14] W. Meng, Y. Liu, Y. Zhu, S. Zhang, D. Pei, Y. Liu, Y. Chen, R. Zhang, S. Tao, P. Sun *et al.*, “Loganomaly: Unsupervised detection of sequential and quantitative anomalies in unstructured logs,” in *IJCAI*, 2019.
- [15] P. He, J. Zhu, S. He, J. Li, and M. R. Lyu, “Tools and benchmarks for automated log parsing,” *International Conference on Software Engineering: Software Engineering in Practice*, 2019.
- [16] J. Soldani and A. Brogi, “Anomaly detection and failure root cause analysis in (micro) service-based cloud applications: A survey,” *ACM Computing Surveys*, vol. 55, no. 3, pp. 1–39, 2022.
- [17] S. Akhtar, S. Khan, and S. Parkinson, “Llm-based event log analysis techniques: A survey,” *arXiv preprint arXiv:2502.00677*, 2025.
- [18] J. Qi, S. Huang, Z. Luan, S. Yang, C. Fung, H. Yang, D. Qian, J. Shang, Z. Xiao, and Z. Wu, “Loggpt: Exploring chatgpt for log-based anomaly detection,” in *IEEE International Conference on High Performance Computing & Communications (HPCC)*, 2023, pp. 273–280.
- [19] W. Guan, J. Cao, S. Qian, J. Gao, and C. Ouyang, “Logllm: Log-based anomaly detection using large language models,” *arXiv preprint arXiv:2411.08561*, 2024.
- [20] Y. Liu, Y. Ji, S. Tao, M. He, W. Meng, S. Zhang, Y. Sun, Y. Xie, B. Chen, and H. Yang, “Loglm: From task-based to instruction-based automated log analysis,” *arXiv preprint arXiv:2410.09352*, 2024.
- [21] M. He, T. Jia, C. Duan, H. Cai, Y. Li, and G. Huang, “Llmelog: An approach for anomaly detection based on llm-enriched log events,” in *IEEE International Symposium on Software Reliability Engineering (ISSRE)*, 2024, pp. 132–143.
- [22] P. Xiao, T. Jia, C. Duan, M. He, W. Hong, X. Yang, Y. Wu, Y. Li, and G. Huang, “Clslog: Collaborating large and small models for log-based anomaly detection,” in *Companion Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering (FSE Companion)*, 2025, pp. 686–690.
- [23] X. Li, Y. Huo, C. Mao, S. Shan, Y. Su, D. Li, and Z. Zheng, “Anomaly-gen: An automated semantic log sequence generation framework with llm for anomaly detection,” *arXiv preprint arXiv:2504.12250*, 2025.
- [24] J. Xu, Z. Cui, Y. Zhao, X. Zhang, S. He, P. He, L. Li, Y. Kang, Q. Lin, Y. Dang *et al.*, “Unilog: Automatic logging via llm and in-context learning,” in *Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (ICSE)*, 2024, pp. 1–12.
- [25] A. Zhong, D. Mo, G. Liu, J. Xie, Y. Chen, Q. Zhang, X. Xu, B. Zhang, W. Wang, Y. Xie *et al.*, “Logparser-llm: Advancing efficient log parsing with large language models,” in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2024, pp. 4559–4570.
- [26] Z. Jiang, J. Liu, Z. Chen, Y. Li, J. Huang, Y. Huo, P. He, J. Gu, and M. R. Lyu, “Lilac: Log parsing using llms with adaptive parsing cache,” *Proceedings of the ACM on Software Engineering (PACMSE)*, vol. 1, 2024.
- [27] S. Song, Y. Zhang, and N. Gao, “Confront insider threat: Precise anomaly detection in behavior logs based on llm fine-tuning,” in *Proceedings of the 31st International Conference on Computational Linguistics (COLING)*, 2025, pp. 8589–8601.
- [28] T. Ahmed, S. Ghosh, C. Bansal, T. Zimmermann, X. Zhang, and S. Rajmohan, “Recommending root-cause and mitigation steps for cloud incidents using large language models,” in *Proceedings of the 45th International Conference on Software Engineering (ICSE)*, 2023, pp. 1737–1749.
- [29] X. Zhang, S. Ghosh, C. Bansal, R. Wang, M. Ma, Y. Kang, and S. Rajmohan, “Automated root causing of cloud incidents using in-context learning with gpt-4,” in *Companion Proceedings of the 32nd ACM International Conference on the Foundations of Software Engineering*, 2024, pp. 266–277.
- [30] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-augmented generation for knowledge-intensive nlp tasks,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474.
- [31] W. Zhang, Q. Zhang, E. Yu, Y. Ren, Y. Meng, M. Qiu, and J. Wang, “Leveraging rag-enhanced large language model for semi-supervised log anomaly detection,” in *IEEE International Symposium on Software Reliability Engineering (ISSRE)*, 2024, pp. 168–179.
- [32] C. Duan, T. Jia, Y. Yang, G. Liu, J. Liu, H. Zhang, Q. Zhou, Y. Li, and G. Huang, “Eagerlog: Active learning enhanced retrieval augmented generation for log-based anomaly detection,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025, pp. 1–5.
- [33] T. Dao, D. Y. Fu, S. Ermon, A. Rudra, and C. Ré, “Flashattention: Fast and memory-efficient exact attention with io-awareness,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 16 344–16 359, 2022.
- [34] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “Lora: Low-rank adaptation of large language models,” *International Conference on Learning Representations*, 2022.
- [35] J. Kaplan, S. McCandlish, T. Henighan, T. B. Brown, B. Chess, R. Child, S. Gray, A. Radford, J. Wu, and D. Amodei, “Scaling laws for neural language models,” *arXiv preprint arXiv:2001.08361*, 2020.
- [36] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt, “Measuring massive multitask language understanding,” *International Conference on Learning Representations*, 2021.
- [37] P. Liang, R. Bommasani, T. Lee, D. Tsipras, D. Soylu, M. Yasunaga, Y. Zhang, D. Narayanan, Y. Wu, A. Kumar *et al.*, “Holistic evaluation of language models,” *Transactions on Machine Learning Research*, 2023.
- [38] C. E. Jimenez, J. Yang, A. Wettig, S. Yao, K. Pei, O. Press, and K. Narasimhan, “Swe-bench: Can language models resolve real-world github issues?” *International Conference on Learning Representations*, 2024.
- [39] T. Z. Zhao, E. Wallace, S. Feng, D. Klein, and S. Singh, “Calibrate before use: Improving few-shot performance of language models,” in *International Conference on Machine Learning (ICML)*, 2021, pp. 12 697–12 706.
- [40] H. Zhou, X. Wan, I. Vulić, and A. Korhonen, “Batch calibration: Rethinking calibration for in-context learning and prompt engineering,” in *International Conference on Learning Representations (ICLR)*, 2024.
- [41] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3982–3992.
- [42] J. Johnson, M. Douze, and H. Jégou, “Billion-scale similarity search with GPUs,” *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2019.
- [43] Qwen Team, “Qwen2.5 technical report,” Alibaba Group, Tech. Rep., 2024.
- [44] A. Q. Jiang, A. Sablayrolles, A. Mensch, C. Bamford, D. S. Chaplot, D. de las Casas, F. Bressand, G. Lengyel, G. Lample, L. Saulnier *et al.*, “Mistral 7b,” *arXiv preprint arXiv:2310.06825*, 2023.
- [45] B. Efron, “Bootstrap methods: Another look at the jackknife,” *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.
- [46] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017, pp. 1321–1330.
- [47] J. Lin, J. Tang, H. Tang, S. Yang, W.-M. Chen, W.-C. Wang, G. Xiao, X. Dang, C. Gan, and S. Han, “Awq: Activation-aware weight quantization for llm compression and acceleration,” *Proceedings of Machine Learning and Systems*, vol. 6, 2024.

- [48] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh, “Gptq: Accurate post-training quantization for generative pre-trained transformers,” *International Conference on Learning Representations*, 2023.
- [49] T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer, “Llm.int8(): 8-bit matrix multiplication for transformers at scale,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 30 318–30 332, 2022.
- [50] H. Touvron, L. Martin, K. Stone, P. Albert, A. Almahairi, Y. Babaei, N. Bashlykov, S. Batra, P. Bhargava, S. Bhosale *et al.*, “Llama 2: Open foundation and fine-tuned chat models,” *arXiv preprint arXiv:2307.09288*, 2023.
- [51] A. Q. Jiang, A. Sablayrolles, A. Roux, A. Mensch, B. Savary, C. Bamford, D. S. Chaplot, D. de las Casas, E. B. Hanna, F. Bressand *et al.*, “Mixtral of experts,” *arXiv preprint arXiv:2401.04088*, 2024.
- [52] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar *et al.*, “Llama: Open and efficient foundation language models,” *arXiv preprint arXiv:2302.13971*, 2023.